# Capacity, unit costs and a supermarket contract

Teacher guidance and answers | Original TBS classroom practice

Use this lesson after students have met capacity, capacity utilisation, unit costs and labour productivity. Students use the calculations to judge a decision, including why a price below unit cost can still add profit, and why the biggest order is not automatically the best. The tasks are planned to take about 55 minutes and have not been trialled in a classroom. The minutes include brief answer checking within each stage but not handing out materials or a full class plenary; allow about 65 minutes in the timetable. Best suited to Year 13 classes who have already covered section 3.4 and are applying or revising it; Year 12 classes starting in September 2026 follow the new 7138 specification.

## Curriculum fit

AQA AS and A-level Business specification (covers 7131 and 7132; this lesson is for A-level 7132), Version 2.0, 14 June 2023, for teaching from September 2023 and A-level exams in 2024 onwards. Section 3.4.2 Analysing operational performance (heading on printed page 17, content on printed page 18): calculate labour productivity, unit costs (average costs), capacity and capacity utilisation. Section 3.4.3: the importance of capacity, including how to utilise capacity efficiently, and difficulties in increasing efficiency and productivity. AQA's 7132 course page says to continue using 7132 for cohorts taking exams in 2027, which AQA's 7138 page lists as the last 7132 exams, and describes the new 7138 specification as first taught from September 2026. Contribution per unit (price minus variable cost) is used in Task 2 as supporting method and is not one of the operations measures listed in 3.4.2. This lesson is NOT mapped to 7138; check that specification before using it with a 7138 class. Checked 8 October 2026. This is a TBS lesson, not an AQA resource, not endorsed by AQA, and uses no AQA exam questions or mark scheme.

All business details and numbers are fictional. The answers are indicative teaching guidance only. There is no official mark tariff, level descriptor or AO allocation for these tasks and no grade claim is made.

## Lesson plan (about 55 minutes of tasks, estimated)

| Stage | Minutes | What happens |
|---|---|---|
| Starter: which measure? | 5 | Match three questions to the three measures. |
| Task 1: the bakery today | 10 | Utilisation, unit cost, productivity, profit. |
| Task 2: why not say no? | 7 | Contribution of a contract loaf versus average cost. |
| Task 3: two options | 15 | Complete the comparison table in pairs. |
| Task 4: test the risk | 6 | Lost regular sales; the tipping point. |
| Task 5: decide | 8 | Short written recommendation. |
| Exit ticket | 4 | Two questions. |
| Total | 55 | |

45-minute version: starter 4, Task 1 8, Task 2 5, Task 3 12, Task 4 4, Task 5 7, exit ticket 5. Everyone completes Option A; give the Option B column of the answer table to the class as data for Task 5; set Task 4b as an extension. Task 5 is tight, so ask for about 60 words or set it as homework. This timing has not been trialled. The 45 minutes are task time only, so in a real 45-minute period, with handing out and feedback, move the exit ticket to homework.

## Answers: starter, Tasks 1 and 2

Indicative guidance for teachers. This is not an AQA mark scheme and carries no marks or grades.

## Support and extension

Support: write the three formulas on the board and complete Task 1a with the class. Extension: Task 4b, and ask what other costs of accepting a supermarket contract the figures leave out (for example payment terms, dependence on one customer and pressure to cut other prices).

## Answers

Starter: (i) labour productivity; (ii) unit cost; (iii) capacity utilisation.

| Task 1 | Answer | Working |
|---|---|---|
| a. Capacity utilisation | 75.0% | 1,500 / 2,000 x 100 |
| b. Total cost; unit cost | £4,200; £2.80 | 3,000 + 0.80 x 1,500 = 4,200; 4,200 / 1,500 |
| c. Labour productivity | 250.0 loaves per baker | 1,500 / 6 |
| d. Weekly profit | £1,200 | 3.60 x 1,500 = 5,400; 5,400 - 4,200 |

Task 2a: one contract loaf adds £2.50 revenue and £0.80 variable cost, so £1.70 extra profit, because fixed costs do not change when the spare capacity is used.

Task 2b: the owner is not right as long as the extra loaves fit within existing capacity and fixed costs do not rise. The £2.80 unit cost includes £2.00 of fixed cost per loaf (3,000 / 1,500) which is paid anyway. Each contract loaf covers its £0.80 variable cost and adds £1.70 towards fixed costs and profit. Accept answers that also note the risks (Task 4) and that this reasoning fails if the contract needed more fixed costs, as in Option B.

## Answers: Task 3 table

The Today column of the student table is already filled in with the Task 1 answers, so students can check their Task 1 work before working out the options.

| | Today | Option A | Option B |
|---|---|---|---|
| Total output (loaves) | 1,500 | 2,000 | 2,300 |
| Capacity (loaves) | 2,000 | 2,000 | 2,800 |
| Capacity utilisation (%) | 75.0 | 100.0 | 82.1 |
| Fixed costs (£) | 3,000 | 3,000 | 4,000 |
| Total cost (£) | 4,200 | 4,600 | 5,840 |
| Unit cost (£ per loaf) | 2.80 | 2.30 | 2.54 |
| Labour productivity (loaves per baker) | 250.0 | 333.3 | 287.5 |
| Weekly profit (£) | 1,200 | 2,050 | 1,560 |

## Working for Task 3 and answers: Task 4

Indicative guidance, not an AQA mark scheme.

Option A: output 1,500 + 500 = 2,000; total cost 3,000 + 0.80 x 2,000 = £4,600; unit cost 4,600 / 2,000 = £2.30; utilisation 100.0%; productivity 2,000 / 6 = 333.3; revenue 3.60 x 1,500 + 2.50 x 500 = 5,400 + 1,250 = £6,650; profit 6,650 - 4,600 = £2,050.

Option B: output 1,500 + 800 = 2,300; fixed costs 3,000 + 1,000 = £4,000; total cost 4,000 + 0.80 x 2,300 = 4,000 + 1,840 = £5,840; unit cost 5,840 / 2,300 = £2.54 (2.5391); utilisation 2,300 / 2,800 = 82.1%; productivity 2,300 / 8 = 287.5; revenue 5,400 + 2.50 x 800 = 5,400 + 2,000 = £7,400; profit 7,400 - 5,840 = £1,560.

Option B sells more loaves but earns £490 less than Option A because the £1,000 second shift adds more cost than the extra 300 contract loaves add profit (300 x £1.70 = £510, less £1,000 extra fixed cost = -£490, relative to A). Check: 2,050 - 1,560 = 490. Option B still beats declining (£1,560 against £1,200).

| Task 4 | Answer | Working |
|---|---|---|
| a. Option A with 100 regular loaves lost | £1,770 | Regular 1,400 x 3.60 = 5,040; contract 500 x 2.50 = 1,250; revenue 6,290; output 1,900; total cost 3,000 + 1,520 = 4,520; profit 1,770 |
| b. Most regular loaves that can be lost | 303 loaves | Contract adds 500 x 1.70 = £850. Each lost regular loaf costs 3.60 - 0.80 = £2.80. 850 / 2.80 = 303.6, so 303 whole loaves still leaves a gain; at 304 the contract is worse than declining (profit £1,198.80). |

Note for 4b: with x regular loaves lost, Option A profit is 2,050 - 2.80x. It stays above £1,200 while x is 303 or fewer and falls to £1,198.80 at 304 lost loaves. The question asks for the largest loss that is still better than declining, so the answer is 303. The accepted answer is 303; if a student answers 304, ask them to check Option A profit at 304 lost loaves (£1,198.80) against £1,200.

## Task 5: indicative decision guidance

A well supported answer chooses one option and uses figures. Example response (not a mark scheme): I recommend Option A. It lifts capacity utilisation from 75.0% to 100.0%, cuts unit cost from £2.80 to £2.30 and raises profit by £850 to £2,050. Option B uses only 82.1% of its larger capacity and earns less (£1,560) at a higher unit cost (£2.54), and declining keeps only £1,200 with a quarter of capacity unused. The main risk is regular customers switching to the supermarket loaf: if more than 303 a week were lost, declining would be better, so I would check sales after a month. Students could reasonably choose B if they argue for long-term growth, or decline if they argue the risk to regular customers or brand is large; they need to use the figures correctly.

Risks and difficulties students may use: regular customers switch to the cheaper loaf; the supermarket may demand lower prices later; running at 100.0% utilisation leaves no slack for breakdowns or rush orders; hiring and training bakers for a second shift is difficult and the contract could end; quality may suffer when ovens run flat out. Look for a risk linked to the case and a figure. Note the simplifications: a real second shift, overtime and a 100% utilisation target would cost more and run less smoothly, and a supermarket contract is usually less generous than this case.

Look for correct figures, a clear comparison of the options, a risk tied to the case and a conclusion that depends on assumptions. Do not award a grade or percentage from these tasks.

## Answers: exit ticket

a. Fixed costs are spread over more loaves, so fixed cost per loaf falls while variable cost per loaf stays at £0.80.

b. Utilisation 1,800 / 2,000 x 100 = 90.0%. Total cost 3,000 + 0.80 x 1,800 = £4,440; unit cost 4,440 / 1,800 = £2.47.

## Sources and limits

Curriculum source: AQA AS and A-level Business specification, Version 2.0, 14 June 2023, section 3.4 (printed pages 17 and 18). The specification verifies the topic scope only. The tasks, figures, timings and guidance are original TBS teaching design.